

5G RAN

Cell Outage Detection and Recovery Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for cell outage detection based on comprehensive features by the UMPThb board. For details, see 3.3.3 Hardware .	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the cell outage detection based on zero access function in NSA networking. For details, see: • 3.1.1 Cell Outage Detection • 3.1.1.5 Based on Zero Access • 3.4.1.1 Activation • 3.4.1.2 Deactivation • 3.4.2 Activation Verification	Modified parameter: Added the NSA_ZERO_ACCESS_DETECT_SW option to the gNBOamParam.CoderAlgoSwitch parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added cell outage detection based on comprehensive features. For details, see: • 2.3 Differences Between NSA and SA • 3.1.1 Cell Outage Detection • 3.1.1.8 Cell Outage Detection Based on Comprehensive Features • 3.1.2 Cell Outage Recovery • 3.3.3 Hardware • 3.4.1.1 Activation • 3.4.1.2 Deactivation • 3.4.2 Activation Verification	Modified parameter: Added the COMPR_FEATURE_DETECT_SW option to the gNBOamParam.CoderAlgoSwitch parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-020101	Reliability	3 Cell Outage Detection and Recovery

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Cell Outage Detection and Recovery	Cell outage detection and recovery functions are supported in both NSA and SA networking. However, the support for cell outage detection based on abnormal preset KPIs and cell outage detection based on comprehensive features differs between NSA and SA networking as follows: • Cell outage detection based on the service setup success rate and cell outage detection based on the abnormal release rate are supported in both NSA and SA networking. • Cell outage detection based on the RRC setup success rate is supported in SA networking only. • Cell outage detection based on comprehensive features is supported only in SA networking.	3 Cell Outage Detection and Recovery

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Cell Outage Detection and Recovery	None	3 Cell Outage Detection and Recovery

3 Cell Outage Detection and Recovery

3.1 Principles

3.1.1 Cell Outage Detection

Cell outage refers to scenarios where UEs cannot access cell services or where there is significant key performance indicator (KPI) deterioration due to faults or alarms. Cell outage severely affects both network performance and user experience, especially on single-frequency and single-RAT networks. Cell outage detection is used throughout the network lifecycle and enables base stations to automatically detect cell outages, which reduces the cell outage detection duration.

Cell outage may be generally caused by:

- Software or hardware faults that can be detected by a gNodeB, including faults in an RRU, baseband processing unit (BBP), common public radio interface (CPRI) port, feeder, power supply system, and transport link (such as an S1 link).
- Unknown faults, including the software or hardware faults that are undetectable by a gNodeB and do not trigger any alarms, such as faults on common or physical channels.

Huawei provides the following cell outage detection mechanisms:

- Cell outage detection based on cell unavailability alarms: The gNodeB monitors its hardware and software status. Once detecting a cell unavailability fault, it reports a correlated alarm, and determines cell outage based on the reported alarm.
- Cell outage detection based on abnormal preset KPIs: The gNodeB determines cell outage based on the three KPIs (service setup success rate, abnormal release rate, and RRC setup success rate).
- Cell outage detection based on the cell sleeping state: The gNodeB periodically determines whether a sleeping cell is in outage based on indicators such as the maximum number of UEs in RRC_CONNECTED mode in a cell and the number of RRC connection setup requests.

- Cell outage detection based on internal auditing: The gNodeB periodically performs internal auditing, helping quickly detect an outage cell.
- Cell outage detection based on zero access: The gNodeB identifies exceptions in the access procedure and determines whether cell outage occurs based on an algorithm for the adaptive exception detection threshold.
- Cell outage detection based on internal faults: The gNodeB determines whether cell outage occurs based on the absence of traffic in a cell, which is caused by internal faults.
- Cell outage detection based on neighboring-cell UE cooperation: The gNodeB determines whether a cell is in outage by verifying the success of incoming handovers of neighboring-cell UEs.
- Cell outage detection based on comprehensive features: The gNodeB determines cell outage based on the comprehensive features of a cell.

The preceding cell outage detection mechanisms include abnormal KPI detection, outage cell detection, and no-traffic cell detection. For details, see [Table 3-1](#).

Table 3-1 Cell outage detection mechanisms

Type	Mechanism	Chapter/Section
Abnormal KPI detection	The detection result indicates that the cell is not completely in outage, KPIs are abnormal, and some UEs cannot access the cell.	3.1.1.2 Based on Abnormal Preset KPIs 3.1.1.4 Based on Internal Auditing
Outage cell detection	The detection result indicates that the cell is completely in outage and all UEs cannot access the cell.	3.1.1.1 Based on Cell Unavailability Alarms 3.1.1.5 Based on Zero Access 3.1.1.6 Based on Internal Faults 3.1.1.7 Cell Outage Detection Based on Neighboring-Cell UE Cooperation
No-traffic cell detection	The detection result cannot be used to determine whether the cell is in outage. It only indicates that the cell has no traffic. If you need to determine cell outage, onsite check is recommended.	3.1.1.3 Based on the Cell Sleeping State

3.1.1.1 Based on Cell Unavailability Alarms

Detection Methods

The gNodeB monitors its hardware and software status. Once detecting a cell unavailability fault, it reports ALM-29841 NR Cell Unavailable to the MAE-Access. After receiving the alarm, the MAE-Access performs one of the following operations:

- If the **Enable cell unavailability fault detection** check box is selected, the MAE-Access considers the cell to be an outage cell due to a cell unavailability fault and adds the cell to the outage cell list for you to query. After the cell unavailability fault is rectified either before or after cell reactivation, the MAE-Access removes this cell from the list and then adds this cell to the list of recovered outage cells.
- If the **Enable cell unavailability fault detection** check box is deselected, the MAE-Access does not determine whether the cell is an outage cell. After the fault is rectified either before or after cell reactivation, the MAE-Access adds the cell to the list of recovered outage cells.

Important Notes

- You are advised to deselect the **Enable cell unavailability fault detection** check box on the MAE-Access before performing any operations on the entire network.

If you perform operations on the entire network with the **Enable cell unavailability fault detection** check box selected, a large number of cells will become outage cells due to cell unavailability faults being reported.

- You are advised to select the **Generate alarm on cell outage** check box on the MAE-Access to effectively process cell outage caused by cell unavailability faults.

Cell unavailability faults significantly affect user experience. To ensure the reliability of cell unavailability fault-based cell outage detection, the MAE-Access can determine cell outage even when the ALM-29841 NR Cell Unavailable alarm is manually masked or is lost due to an alarm storm.

However, outage cell data cannot be obtained from the alarm system in cases where the alarm is masked. Therefore, you are advised to enable alarm generation by selecting the **Generate alarm on cell outage** check box on the MAE-Access following instructions provided in [3.4.2 Activation Verification](#).

If you deactivate the outage cell caused by cell unavailability faults, ALM-29841 NR Cell Unavailable will be cleared. However, if such alarms caused by cell unavailability faults persist, the outage cell is not added to the list of recovered outage cells and remains in the outage cell list.

3.1.1.2 Based on Abnormal Preset KPIs

The gNodeB determines whether a cell is an outage cell based on the KPIs service setup success rate, abnormal release rate, and RRC setup success rate. Cell outage detection based on the RRC setup success rate applies only to SA networking. Cell outage detection based on the service setup success rate and cell outage detection based on the abnormal release rate apply to both NSA networking and SA networking. The gNodeB tracks the values of these KPIs for the cell over

consecutive performance measurement periods to identify cell outage. This feature cannot identify cell outage based on KPI changes caused by UE faults or coverage problems, such as weak coverage or coverage holes. If any KPI for a cell returns an unexpected value, the gNodeB determines that the cell is an outage cell and notifies the MAE-Access of the outage cell. [Table 3-2](#) describes the criteria for cell outage detection based on abnormal preset KPIs. The MAE-Access considers the cell recovered only after KPIs return to normal.

 NOTE

The measurement period is the period by which cell outage detection based on abnormal preset KPIs is performed. For details about checking the measurement periods, see [Measurement Period Check](#).

Table 3-2 Criteria for determining cell outages based on abnormal preset KPIs

KPI	Criteria
Service setup success rate	Service setup success rate < Service setup success rate(%) when decisions are made twice in a row. A decision is made in a cumulative period when the cumulative total of service setup requests in a cell exceeds the threshold Service setup request number .
Abnormal release rate	Abnormal release rate ≥ Abnormal release rate(%) when decisions are made twice in a row. A decision is made in a cumulative period when the cumulative total of service releases in a cell exceeds the threshold Service release number . Each abnormal service release is considered a call drop.
RRC setup success rate	RRC setup success rate < RRC setup success rate(%) when decisions are made twice in a row. A decision is made in a cumulative period when the cumulative total of RRC connection setup requests in a cell exceeds the threshold RRC setup request number .
Using the service setup success rate as an example, if the cumulative total of service setup requests in a cell does not exceed the threshold Service setup request number within a measurement period, the cumulative total is added to the measurement result of the service setup request number in the next measurement period. This process is repeated until the cumulative total exceeds the corresponding threshold at the end of a measurement period. Then, a decision is made, and the cumulative total of Service setup request number is reset to zero. The maximum length of a cumulative period is 24 hours.	

[Table 3-3](#) describes the preset KPI thresholds.

Table 3-3 Preset KPI thresholds

Preset KPI Threshold	GUI Value Range	Default Value	Description
Service setup request number	1 to 4294967295	20	Used for detecting outage cells based on abnormal preset KPIs. Users can set the thresholds for related KPIs on the MAE-Access. If any of these KPIs for a cell does not meet the requirements for their respective thresholds, the MAE-Access considers this cell to be an outage cell.
Service setup success rate(%)	1 to 100	5	
Service release number	1 to 4294967295	20	
Abnormal release rate(%)	1 to 100	95	
RRC setup request number	1 to 4294967295	20	
RRC setup success rate(%)	1 to 100	5	

Table 3-4 and **Table 3-5** describe the counters used in the following formulas for calculating these KPIs.

In NSA networking,

- Service setup success rate = Number of successful DRB additions for LTE-NR NSA DC UEs on the SgNB/Number of DRB addition requests for LTE-NR NSA DC UEs on the SgNB
- Abnormal release rate = Number of abnormal DRB releases for LTE-NR NSA DC UEs on the SgNB/Number of DRB releases for LTE-NR NSA DC UEs on the SgNB

Where,

- The number of DRB addition requests for LTE-NR NSA DC UEs on the SgNB indicates the cumulative total of **Service setup request number**.
- The number of DRB releases for LTE-NR NSA DC UEs on the SgNB indicates the cumulative total of **Service release number**.

In SA networking,

- Service setup success rate = Number of successful QoS flow setups/Number of QoS flow setup attempts

- Abnormal release rate = Number of abnormal QoS flow releases/(Number of abnormal QoS flow releases + Number of normal QoS flow releases)
- RRC setup success rate = Number of successful RRC connection setups/
Number of RRC connection setup requests

Where,

- The number of QoS flow setup attempts indicates the cumulative total of **Service setup request number**.
- The sum of the number of abnormal QoS flow releases and the number of normal QoS flow releases indicates the cumulative total of **Service release number**.
- The number of RRC connection setup requests indicates the cumulative total of **RRC setup request number**.

Table 3-4 Counters in NSA networking

Counter ID	Counter Name	Counter Description
1911816754	N.NsaDc.DRB.Add.Att	Number of DRB addition requests for LTE-NR NSA DC UEs on the SgNB
1911816755	N.NsaDc.DRB.Add.Succ	Number of successful DRB additions for LTE-NR NSA DC UEs on the SgNB
1911816756	N.NsaDc.DRB.Rel	Number of DRB releases for LTE-NR NSA DC UEs on the SgNB
1911816757	N.NsaDc.DRB.AbnormRel	Number of abnormal DRB releases for LTE-NR NSA DC UEs on the SgNB

Table 3-5 Counters in SA networking

Counter ID	Counter Name	Counter Description
1911816962	N.QosFlow.Est.Att	Number of QoS flow setup attempts
1911816963	N.QosFlow.Est.Succ	Number of successful QoS flow setups
1911816966	N.QosFlow.NormRel	Number of normal QoS flow releases
1911816967	N.QosFlow.AbnormRel	Number of abnormal QoS flow releases
1911816950	N.RRC.SetupReq.Succ	Number of successful RRC connection setups
1911816954	N.RRC.SetupReq.Att	Number of RRC connection setup requests

If the KPI that results in cell outage meets the criteria described in [Table 3-6](#) before or after cell reactivation, the MAE-Access considers that the outage cell has

recovered and removes it from the outage cell list. [Table 3-6](#) lists the criteria for cell outage recovery detection based on abnormal preset KPIs.

Table 3-6 Criteria for determining cell outage recovery based on abnormal preset KPIs

KPI	Criteria
Service setup success rate	The service setup success rate is greater than or equal to the threshold Service setup success rate(%) plus 5% (or 100% when Service setup success rate(%) plus 5% is greater than 100%) in one decision period. A decision is made in a cumulative period when the cumulative total of service setup requests in a cell exceeds the threshold Service setup request number .
Abnormal release rate	The abnormal release rate is less than the threshold Abnormal release rate(%) minus 5% (or 1% when Abnormal release rate(%) minus 5% is less than 1%) in one decision period. A decision is made in a cumulative period when the cumulative total of service releases in a cell exceeds the threshold Service release number .
RRC setup success rate	The RRC setup success rate is greater than or equal to the threshold RRC setup success rate(%) plus 5% (or 100% when RRC setup success rate(%) plus 5% is greater than 100%) in one decision period. A decision is made in a cumulative period when the cumulative total of RRC connection setup attempts in a cell is greater than the threshold RRC setup number .

Cell outage detection based on abnormal KPIs listed in [Table 3-2](#) does not apply to the following scenarios:

- During energy conservation and emission reduction, the cell is in the carrier shutdown state.
- The cell is blocked.
- No resources are available in a cell due to cell overload. In this case, cell outage detection based on the service setup success rate does not apply.
- The cell is barred. In this case, UEs cannot initiate an RRC connection setup in this cell and therefore cell outage detection based on the RRC setup success rate does not apply.
- The **Enable preset KPI detection** check box is deselected.

The MAE-Access adds an outage cell detected based on an abnormal KPI to the outage cell list. Users can view this list on the MAE-Access.

Pay attention to the following:

- An outage cell may recover in the next performance measurement period after being detected, and such a scenario cannot be avoided even if this feature is used. This is more likely to occur if users set the threshold **Service setup success rate(%)** or **RRC setup success rate(%)** to a value greater than the default value, or set the threshold **Abnormal release rate(%)** to a value

less than the default value. To reduce this probability as much as possible, it is recommended that you set the thresholds **Service setup request number**, **Abnormal release number**, and **RRC setup request number** to 60 or above, especially during initial network deployment where weak coverage issues or coverage holes are more likely to occur.

- If any of the optical module-related alarms exists, and the alarm severity is major, "The current cell in outage has an optical module fault. Check it preferentially." will be displayed in the cell outage cause. The optical module-related alarms include ALM-26507 RF Unit Optical Module Fault, ALM-26504 RF Unit CPRI Interface Error, and ALM-26234 BBU CPRI Interface Error.

3.1.1.3 Based on the Cell Sleeping State

Criteria for Cell Outage and Cell Recovery Detection

A sleeping cell is a cell that generates no alarms when potential gNodeB software or hardware faults cause a high probability of UE access failures and therefore the cell carries no traffic for a long period.

The gNodeB checks whether a sleeping cell is in outage based on the no-traffic duration or abnormal incoming handovers during a specified monitoring period (08:00:00 to 20:00:00 every day). The specific criteria are described in [Table 3-7](#).

Table 3-7 Criteria for determining cell outages based on the cell sleeping state

Scenario	Criteria
Measuring no-traffic duration	Enable long-time no traffic judgment is selected and the following conditions apply for a cell within N consecutive hours (N is specified by the No-traffic period threshold (h) parameter. For details about the parameter setting, see Table 3-16): • The N.RRC.SetupReq.Att counter returns value 0. • The N.User.RRCCConn.Max counter returns value 0.
Detecting abnormal incoming handovers	Enable long-time no traffic judgment is selected and the following conditions apply for a cell within a certain measurement period: • The N.RRC.SetupReq.Att counter returns value 0. • The N.User.RRCCConn.Max counter returns value 0. • The N.HO.IntragNB.PrepAttIn counter returns a value other than 0, and the sum of its values in this measurement period and in the previous measurement period is greater than a fixed threshold (10; not configurable).

Table 3-8 Related counters

Counter ID	Counter Name	Counter Description
1911816954	N.RRC.SetupReq.Att	Number of RRC connection setup requests

Counter ID	Counter Name	Counter Description
1911816772	N.User.RRCCConn.Max	Maximum number of UEs in RRC_CONNECTED mode in a cell
1911816906	N.HO.IntragNB.PrepAttn	Number of intra-gNodeB incoming handover attempts

A cell is considered to be in outage when any scenario with specific criteria described in [Table 3-7](#) applies. If a cell is in outage but none of the criteria for determining cell outage described in [Table 3-7](#) applies, this cell recovers.

- Upon detecting a cell in outage, the gNodeB reports a cell outage event to the MAE-Access. The MAE-Access then adds this cell to the outage cell list for user query.
- Upon detecting that an outage cell recovers, the gNodeB reports a cell outage recovery event to the MAE-Access. The MAE-Access then removes this cell from the outage cell list and adds this cell to the list of recovered outage cells.

The cell outage detection criteria based on the cell sleeping state do not apply to the following scenarios:

- UEs in a barred cell are being prevented from initiating RRC connection setup requests. In this scenario, the gNodeB does not determine whether the cell is in outage based on the value of the **N.RRC.SetupReq.Att** counter.
- The cell becomes unavailable due to manual intervention, system faults, or energy saving reasons.
- The cell is blocked.
- The current time is beyond the monitoring period.

Optimization for High CPRI BER Scenarios

Due to the high BER on the CPRI link, the RF module discards all data. As a result, the gNodeB cannot identify the CPRI transmission fault and considers that the cell is a sleeping cell. To address this issue, the **HIGHCPRIBER_FAULT_POSTPROC_SW** option of the **Equipment.EqmOptFeatSw** parameter is added to specify whether to enable optimization for high CPRI BER scenarios.

When this function is enabled, if the RF module discards all data due to a high BER on the CPRI link, the BBU sets the status of the cell corresponding to the RF module to unavailable and reports ALM-29870 NR DU Cell TRP Unavailable or ALM-29871 NR DU Cell TRP Capability Degraded with "Specific Problem" being "CPRI interface abnormal".

3.1.1.4 Based on Internal Auditing

The gNodeB performs an internal auditing based on counters every five minutes and checks whether the counter values in three consecutive periods comply with the preset exception formula for effectively determining outage cells.

As listed in [Table 3-9](#), if the gNodeB determines that a cell is in outage based on internal auditing, it reports the outage cell to the MAE-Access.

Table 3-9 Criteria for determining cell outages based on internal auditing

Scenario	Criteria
Random access	A cell is considered to be in outage if the following conditions are met in three consecutive periods for internal auditing: • The number of preambles (for contention-based and non-contention-based random access only) exceeds the specified threshold 20. • The number of preambles is more than two times that of random access responses (RARs); that is, the number of RARs is less than half of the number of preambles.
RRC connection setup	A cell is considered to be in outage if the following conditions are met in three consecutive periods for internal auditing: • The number of baseband resource preparation requests (triggered by RRC connection setup and RRC connection reestablishment) exceeds the specified threshold 20. • The number of baseband resource preparation requests is more than two times that of baseband resource preparation successes; that is, the number of baseband resource preparation successes is less than half of the number of baseband resource preparation requests.

The MAE-Access adds an outage cell whose gNodeB internal auditing result is abnormal to the outage cell list. Users can view this list on the MAE-Access. The cell is considered to be in outage before the gNodeB internal auditing results indicate the cell returns to normal.

After an outage cell meets the recovery conditions listed in [Table 3-10](#), the cell is considered to recover, and the MAE-Access removes it from the outage cell list.

Table 3-10 Criteria for outage cell recovery

Scenario	Criteria
Random access	A cell is considered to be recovered if the following conditions are met in three consecutive periods: • The number of preambles (for contention-based and non-contention-based random access only) exceeds the specified threshold 20. • The number of preambles is no more than two times that of RARs.

Scenario	Criteria
RRC connection setup	A cell is considered to be recovered if the following conditions are met in three consecutive periods: - The number of baseband resource preparation requests (triggered by RRC connection setup and RRC connection reestablishment) exceeds the specified threshold 20. - The number of baseband resource preparation requests is no more than two times that of baseband resource preparation successes.

3.1.1.5 Based on Zero Access

Cell outage detection based on zero access supports both SA networking ([Cell Outage Detection Based on Zero Access in SA Networking](#)) and NSA networking ([Cell Outage Detection Based on Zero Access in NSA Networking](#)).

Cell Outage Detection Based on Zero Access in SA Networking

The gNodeB detects whether cell outage occurs by identifying exceptions in the random access and RRC connection setup procedures. This function is controlled by the **ZERO_ACCESS_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter and applies to SA networking as well as NSA and SA hybrid networking.

When this option is selected, the function of cell outage detection based on zero access takes effect for the gNodeB. If the criteria described in [Table 3-11](#) are met, the gNodeB determines that the cell is in outage. The MAE-Access adds the outage cell to the outage cell list. Users can view this list on the MAE-Access. The cell is considered to be in outage before the gNodeB considers the cell returns to normal.

Table 3-11 Criteria for determining zero access

Scenario	Criteria
There is Msg1 but no Msg2.	The gNodeB determines that the cell is in outage if all the following conditions are met for one consecutive hour during busy hours ^(a): - The number of Msg1 messages for contention-based random accesses (N.RA.Contention.Att) is greater than the Msg1 threshold for cell outage detection based on zero access. Herein, the Msg1 threshold is automatically calculated by the gNodeB based on the number of Msg1 messages during busy hours. - Number of Msg2 messages for contention-based random access responses (N.RA.Contention.Resp) = 0 - Duration of cell availability $\geq 3595s$

Scenario	Criteria
There is Msg2 but no Msg3.	During busy hours, the gNodeB checks every hour whether cell outage occurs. If the number of consecutive hours during which the following conditions are met reaches the value of zero access period number threshold (set on the MAE), the gNodeB determines that the cell is in outage. • The number of Msg2 messages for contention-based random access responses (N.RA.Contention.Resp) is greater than the Msg2 threshold for cell outage detection based on zero access. Herein, the Msg2 threshold is automatically calculated by the gNodeB based on the number of Msg2 messages during busy hours (N.RA.Contention.Resp). • Number of Msg3 messages for RRC connection setup requests (N.RRC.SetupReq.Att) = 0 • Duration of cell availability $\geq 3595s$ • Number of Msg3 messages during non-contention-based incoming handovers (N.RA.Dedicated.HO.Msg3) < 10 • Cell uplink interference < -95 dBm
There is Msg3 but no Msg5.	The gNodeB determines that the cell is in outage if all the following conditions are met for one consecutive hour during busy hours: • The number of Msg3 messages for RRC connection setup requests (N.RRC.SetupReq.Att) is greater than the Msg3 threshold for cell outage detection based on zero access. Herein, the Msg3 threshold is automatically calculated by the gNodeB based on the number of Msg3 messages during busy hours (N.RRC.SetupReq.Att). • Number of Msg5 messages for successful RRC connection setups (N.RRC.SetupReq.Succ) = 0 • Duration of cell availability $\geq 3595s$
a: Busy hours are automatically calculated by the gNodeB based on N.PDCP.Vol.UL.TrfSDU.Tx and N.PDCP.Vol.DL.TrfSDU.Rx .	

When a cell is in outage, the gNodeB checks the number of Msg5 messages for successful RRC connection setups (**N.RRC.SetupReq.Succ**) in real time. If the number is greater than 0, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

Cell Outage Detection Based on Zero Access in NSA Networking

The gNodeB identifies outage cells by detecting SgNB addition exceptions. This function is controlled by the **NSA_ZERO_ACCESS_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter and applies to NSA networking.

When this option is selected, the function of cell outage detection based on zero access takes effect for the gNodeB. If the criteria described in [Table 3-12](#) are met, the gNodeB determines that the cell is in outage. The MAE-Access adds the

outage cell to the outage cell list. Users can view this list on the MAE-Access. The cell is considered to be in outage before the gNodeB considers the cell returns to normal.

Table 3-12 Criteria for determining zero access

Scenario	Criteria
There are SgNB addition attempts but the number of SgNB addition acknowledgements is 0.	The gNodeB determines that the cell is in outage if all the following conditions are met for one consecutive hour during busy hours ^(a): • Number of SgNB addition attempts (indicated by the result of N.NsaDc.DRB.Add.Att + N.NsaDc.InterSgNB.SgNB.Add.Att) > a specific threshold (calculated by the gNodeB based on the number of SgNB addition attempts during busy hours in the cell) • Number of SgNB addition acknowledgements (N.NsaDc.SgNB.Add.Ack + N.NsaDc.InterSgNB.SgNB.Add.Ack) = 0 • Duration of cell availability ≥ 3595s
a: Busy hours are automatically calculated by the gNodeB based on N.PDCP.Vol.UL.TrfSDU.Tx and N.PDCP.Vol.DL.TrfSDU.Rx .	

When a cell is in outage, the gNodeB checks the number of SgNB addition acknowledgements (**N.NsaDc.SgNB.Add.Ack** + **N.NsaDc.InterSgNB.SgNB.Add.Ack**) in real time. If the number is greater than 0, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

3.1.1.6 Based on Internal Faults

If there is no Msg1 message, cell outage detection based on abnormal KPIs cannot be used because it cannot be determined whether cell outage is caused by device faults or no UE access. Therefore, the function of cell outage detection based on internal faults is introduced. This function is controlled by the **INTERNAL_FAULT_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter.

This function detects factors such as TX/RX power, links, and resources. In the scenario where no Msg1 message exists, if the historical status of related devices is normal and there are UEs performing services, but after a certain time, the device status is abnormal and no UE is performing services, then the gNodeB determines that no UE is active due to abnormal device status, causing the cell to enter the sleeping state. In this case, the gNodeB reports ALM-29875 No Traffic on NR DU Cell TRP and reports sleeping cells to the MAE CODR module. The fault duration threshold for triggering alarm reporting is specified by the **gNBOamParam.SlcAlmDetectDurationThld** parameter and the recommended value is 24.

3.1.1.7 Cell Outage Detection Based on Neighboring-Cell UE Cooperation

For a cell without Msg1 for a long time, it cannot be directly classified as an outage cell because it is unclear whether the outage is due to no UE access or UE access failure. These cells are referred to as suspected outage cells. To accurately identify suspected outage cells as outage cells and increase the detection coverage, the function of cell outage detection based on neighboring-cell UE cooperation is introduced. This function is controlled by the **NCELL_UE_COOP_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter.

After this function is enabled, the identification process is as follows:

1. Based on the counters related to incoming handovers between specific neighboring cells (such as **N.HO.NCell.ExecSuccln**), the base station selects six intra-frequency neighboring cells with the greatest number of historical handovers to a suspected outage cell.
2. From each selected neighboring cell, the base station selects a UE that can measure the downlink signals of the suspected outage cell and the UE attempts to hand over to the suspected outage cell.
3. If there are three or more neighboring-cell UEs that attempt to hand over to the suspected outage cell but all the attempts fail, the suspected outage cell is considered as an outage cell.

When a cell is in outage, the gNodeB checks the number of successful handovers of UEs from neighboring cells to the outage cell every 5 minutes. If one UE is successfully handed over to the outage cell, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

3.1.1.8 Cell Outage Detection Based on Comprehensive Features

Cell outage detection based on comprehensive features aims to improve the accuracy of identifying an outage cell when the cell meets both of the following conditions: (1) there is no Msg3 message reported during a sustained period; (2) the criteria described in [3.1.1.5 Based on Zero Access](#) do not apply. This function is controlled by the **COMPR_FEATURE_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter and applies to SA networking.

When cell outage detection based on comprehensive features is enabled, the gNodeB determines cell outage as follows:

1. The gNodeB filters ranges of hours during which a cell remains fully available across five consecutive days. (A cell is considered fully available if it is available for at least 3,595s per hour.) If the number of Msg3 messages (**N.RRC.SetupReq.Att**, representing the number of RRC connection setup requests) is 0 in the cell during the ranges of filtered hours, the cell is considered a suspected outage cell.
2. The gNodeB further determines whether a suspected outage cell is an outage cell based on the features listed in [Table 3-13](#).

Table 3-13 Feature items used for determination of an outage cell

Feature Item	Related Counter
System information	N.SI.Uu.Num
Cell uplink interference	N.UL.NI.Avg
Average downlink transmit power in a cell	N.DL.Pwr.Avg
Uplink sector RSSI	N.UL.RSSI.Avg
Downlink transmit power of RF units	VS.RADIOEQM.DL.Pwr.Avg
Number of seconds when CPRI port detection results are invalid	max(VS.RADIOEQM.CPRI0.UIVS, VS.RADIOEQM.CPRI0.DIVS, VS.RADIOEQM.CPRI1.UIVS, VS.RADIOEQM.CPRI1.DIVS)
Duration of successful channel calibration	N.TRP.Succ.Calib.Dur

The outage status of an outage cell is reported to the CODR module of the MAE-Access, with an outage reason of "Internal fault cell outage."

The gNodeB checks the number of Msg5 messages (**N.RRC.SetupReq.Succ**) every 5 minutes in the outage cell. If the number is greater than 0, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

3.1.2 Cell Outage Recovery

Recovering Outage Cells

Three cell outage recovery modes are available:

- **Recover After Manual Acknowledgement:** The system waits for the manual selection of a recovery method before recovering an outage cell. The following recovery methods are involved: **Reactivate Cell**, **Reset Base**

Station, Reset RRU, Reset RRU Through Powering-off, Reset BBP, and Reset BBP Through Powering-off.

- **Auto Recover:** The system automatically attempts to recover an outage cell. Only one recovery method is available: **Reactivate Cell**. If an outage cell is detected based on an alarm, **Auto Recover** does not take effect.
- **No Recover:** The system does not recover outage cells. The cells wait to be recovered using other methods such as manual cell reconfiguration and cell reactivation.

In **Auto Recover**, the MAE-Access performs only cell reactivation. In this process, the gNodeB reconfigures cell data and requests cell resources again.

The MAE-Access provides automatic and manual cell reactivation methods. After an outage cell is reactivated, the cell status is displayed as **running** on the MAE-Access. The MAE-Access then continues monitoring the cell.

Table 3-14 describes the policies for handling different types of cell outage.

Table 3-14 Cell outage types and handling policies

Cell Outage Type	Cell Outage Detection Method	Cell Outage Handling Policy
Cell outage detection based on cell unavailability alarms	3.1.1.1 Based on Cell Unavailability Alarms	If ALM-29841 NR Cell Unavailable is displayed on the MAE-Access, you are advised to troubleshoot the fault and clear the alarm based on the alarm information. After the alarm is cleared, check whether the outage cell recovers.
Cell outage detection based on abnormal preset KPIs	3.1.1.2 Based on Abnormal Preset KPIs	If preset KPIs return unexpected values, you are advised to observe the KPIs during the most recent 72 or more hours within the cell outage period and check the KPI failure causes. If the KPIs were confirmed as abnormal, it is recommended that the cell be recovered as soon as possible.

Cell Outage Type	Cell Outage Detection Method	Cell Outage Handling Policy
Cell outage detected based on zero traffic for a long time	3.1.1.3 Based on the Cell Sleeping State	If a cell is determined as a sleeping cell (outage cell) due to zero traffic for a long time, obtain and observe the KPIs during the most recent 72 or more hours, including the cell outage period. • If the historical traffic of a cell changes significantly at the time around cell outage (from active service access to no service access), the cell is probably a sleeping cell. In this situation, it is recommended that the cell be recovered. • If the historical traffic of a cell fluctuates sharply, it is recommended that cell recovery be postponed. About 2 to 3 hours later, confirm the cell status again. If the cell outage persists, it is recommended that the cell be recovered. If there is no traffic in the cell for a long time, users cannot exactly determine that the cell is a sleeping cell. This is because the historical traffic and the number of UEs fluctuate greatly during the most recent 72 or more hours. In this situation, it is recommended that dialing tests be performed in the cell center to check whether service access is normal in the cell. If the dialing test results show that the service access is abnormal, it is recommended that the cell be recovered as soon as possible.
Cell outage detected based on internal auditing	3.1.1.4 Based on Internal Auditing	If a cell is determined as an outage cell based on auditing results, it is recommended that the KPIs related to cell outage caused by either the sleeping state or KPI exceptions be checked. • If KPIs fluctuate unexpectedly in a cell, for example, the RRC connection setup success rate decreases sharply in the cell or the cell suddenly changes from active service access to no service access, it is recommended that the cell be recovered as soon as possible. • If no KPIs fluctuate unexpectedly in a cell, it is recommended that the cell status be confirmed again at least one hour later. If the cell outage persists, the KPIs need to be checked again. If KPIs fluctuate unexpectedly, it is recommended that the cell be recovered. Otherwise, you are advised to contact Huawei technical support to locate and rectify the fault.

Cell Outage Type	Cell Outage Detection Method	Cell Outage Handling Policy
Cell outage caused by abnormal access	3.1.1.5 Based on Zero Access 3.1.1.6 Based on Internal Faults 3.1.1.7 Cell Outage Detection Based on Neighboring-Cell UE Cooperation 3.1.1.8 Cell Outage Detection Based on Comprehensive Features	If a cell is in outage, it is recommended that the cell be recovered as soon as possible. If, after waiting at least one hour, the cell outage persists after recovery operations are performed or the onsite test result shows that UEs cannot access the cell, you are advised to contact Huawei technical support to locate and rectify the fault.

Self-Healing Policies for Board Component Failures

To achieve precise self-healing and alarm reporting, the **BRDCOMPEXCPROPOL** MO can be used to configure the policies for handling board component failures. The base station will continuously detect the failures of components such as the DDR or cache, and perform self-healing for board component failures only when the cell falls into outage types in [Table 3-15](#).

Table 3-15 Cell outage types

Cell Outage Type	Cell Outage Detection Mode with Board Component Self-Healing
Cell outage caused by abnormal access	3.1.1.5 Based on Zero Access 3.1.1.6 Based on Internal Faults 3.1.1.7 Cell Outage Detection Based on Neighboring-Cell UE Cooperation
Cell outage caused by baseband unit exceptions ^a	The baseband processing unit exception notification function is used to notify the device management module that the baseband unit is abnormal. This function is controlled by the BBP_ABN_NOTIFY_SW option of the gNBOamParam.DevAbnCellUnavblNotifySw parameter.
a: A cell must be bound to a unique baseband processing unit.	

The configuration requirements for self-healing policies for board component failures are as follows:

- Only the baseband processing unit or main control board supports self-healing. If the **MPT** MO is configured in the slave subrack, the **BRDCOMPEXCPROPOL** MO cannot be configured for the board.
- Only one **BRDCOMPEXCPROPOL** MO can be configured for a self-healing object.
- The **BRDCOMPEXCPROPOL.ASSOABNRANGE** parameter specifies the abnormal association range. If the self-healing object is a main control board, **BRDCOMPEXCPROPOL.ASSOABNRANGE** can only be set to **ALL_RAT**.

When the **BRDCOMPEXCPROPOL.ASSOABNRANGE** parameter is set to different values, the impact on cell services during the self-healing period is as follows:

- **SINGLE_CELL**: If there are other cells configured on the board, services in these cells are unavailable during board self-healing.
- **SINGLE_RAT_ALL_CELL**: If there are other RATs on the board, services in the cells of these RATs are unavailable during board self-healing.

Self-healing policies for board component failures can be executed no more than once a week and three times a month. If this limit is exceeded, self-healing is not performed. Additionally, the number of self-healing times is not cleared after the board component failure handling policy is changed. After the self-healing is implemented, components—such as DDR or cache—are checked again. If a component fault persists, ALM-26200 Board Hardware Fault is reported.

3.2 Network Analysis

3.2.1 Benefits

Cell outage detection enables gNodeBs to quickly detect cell outage and shorten the cell outage duration. It also minimizes the impact of cell outage on user experience and helps improve operators' brand image through cell outage self-recovery.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

Cell outage detection allows operators to quickly detect outage cells. In the best-case scenario, cell outage with abnormal KPIs can be detected within 30 minutes. This helps minimize the duration within which cell outage affects network services and KPIs.

When the "cell outage detection based on neighboring-cell UE cooperation" function is enabled, the **Intra-Frequency Handover Out Success Rate (CU)** of the coordinating neighboring cells may slightly deteriorate as UEs in those cells perform NR cell global identifier (NCGI) measurements and coverage-based handovers.

3.3 Requirements

3.3.1 Licenses

None

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NR cell outage detection and recovery	None	Cell outage detection and recovery and license-related information in <i>iMaster MAE Product Documentation</i>	Before enabling the cell outage detection and recovery function on the NE side, ensure that the SON > Cell Outage Detection and Recovery > NR Cell Outage Detection and Recovery menu is available on the MAE-Access. (The license for MMFD-201100 MBB Network Access Package must be loaded.) If the menu is still unavailable after the license is loaded, contact Huawei technical support.
Low-frequency TDD High-frequency TDD	Cell outage detection based on zero access	ZERO_ACCESS_DETECT_SW option of the gNB0amParam.CodrAlgoSwitch parameter	<i>Cell Outage Detection and Recovery</i>	Before enabling the cell outage detection based on neighboring-cell UE cooperation function, you must enable the cell outage detection based on zero access function.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency neighboring cell statistics	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	The cell outage detection based on neighboring-cell UE cooperation function takes effect only when the intra-frequency neighboring cell statistics function is enabled.
Low-frequency TDD High-frequency TDD	Baseband resource mutual aid	NRDUCellTrp.BbResMutualAidSw	None	The baseband processing unit exception notification function can be enabled only when all of the following conditions are met: • The NRDUCellTrp.BasebandEqmId parameter is not set to 255 for all TRPs of an NR DU cell. • The NRDUCellTrp.SecondaryBasebandEqmId parameter is set to 255 for all TRPs of an NR DU cell. • The NRDUCellTrp.BbResMutualAidSw parameter is not set to ON for all TRPs of an NR DU cell.
Low-frequency TDD High-frequency TDD	Secondary Baseband Equipment ID	NRDUCellTrp.SecondaryBasebandEqmId	None	
Low-frequency TDD High-frequency TDD	Baseband Equipment ID	NRDUCellTrp.BasebandEqmId	None	

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

None

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The function of cell outage detection based on internal faults applies only to 3900 and 5900 series base stations (macro base stations).

Other functions are supported by the following base station models:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For cell outage detection based on comprehensive features:
 - Baseband processing units: All NR-capable baseband processing units can be used.
 - Main control board: All NR-capable UMPTE, UMPTg, UMPTga, UMPThb, and UMPTha series boards can be used. In addition, UMPTE series boards must work in NR-only mode.
- For self-healing policies for board component failures:
 - Baseband processing units: All NR-capable UBBPg and UBBPh series boards can be used.
 - Main control boards: All NR-capable UMPTE and UMPTg series boards can be used.
- For other functions:

All NR-capable main control boards and baseband processing units can be used. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Optimization for high CPRI BER scenarios: RF modules whose CPRI port protocol type is eCPRI or IR and 3000 series pRRU modules do not support this function.

The following are the RF modules that support functions other than optimization for high CPRI BER scenarios.

All NR-capable RF modules can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

3.3.4 Others

None

3.3.5 Cells

The function of cell outage detection based on internal faults has the following cell requirements:

- Cell type: low-frequency common cells
(**NRDUCell.NrDuCellNetworkingMode** set to **NORMAL_CELL**)
- Power configuration:
Low-frequency TDD: (**NRDUCellTrp.MaxTransmitPower** +
NRDUCellTrpBeam.MaxSsbPwrOffset × 10) ≥ 300 (0.1 dBm)

 NOTE

If **NRDUCellTrp.PowerConfigMode** is set to **TRANSMIT_POWER_MW**, **NRDUCellTrp.MaxTransmitPower** in the preceding formulas must be replaced with **NRDUCellTrp.MaxTransmitPowerMw**, and then the unit needs to be converted from mW to 0.1 dBm.

Other functions have no cell requirements.

3.4 Operation and Maintenance

3.4.1 Data Configuration

3.4.1.1 Activation

Cell Outage Detection and Recovery

Perform [Step 1](#) through [Step 3](#) for the following types of cell outage detection:

- Cell outage detection based on cell unavailability alarms
- Cell outage detection based on abnormal preset KPIs
- Cell outage detection based on the cell sleeping state
- Cell outage detection based on internal auditing
- Cell outage detection based on zero access
- Cell outage detection based on internal faults
- Cell outage detection based on neighboring-cell UE cooperation
- Cell outage detection based on comprehensive features

Step 1 On the MAE-Access, choose **MAE-Access > SON > NR Cell Outage Detection and Recovery**. The cell outage management window is displayed.

Step 2 Choose **Parameter Settings > Base Station Monitoring**. In the displayed dialog box, select gNodeBs where cell outage detection is to be enabled, and then click **Apply**.

Step 3 Set cell outage detection parameters.

1. Click **Basic Parameters**. A dialog box for parameter settings is displayed. [Table 3-16](#) describes the parameters settings for this function.

Table 3-16 Basic parameter descriptions

Parameter Name	Description
Enable preset KPI detection	If this option is selected, outage cells can be detected based on abnormal preset KPIs.
Enable internal audit	If this option is selected, outage cells can be detected based on internal auditing.
Enable cell unavailability fault detection	If this option is selected, outage cells can be detected based on cell unavailability faults.
Enable zero access detection	If this option is selected, outage cells can be detected based on zero access.
Zero access period number threshold (h)	This parameter specifies the threshold of the number of periods for cell outage detection based on zero access. The default value is 12 in unit of hour.
Enable internal fault detection	If this option is selected, outage cells can be detected based on internal faults or comprehensive features.
Enable neighboring cell UE coordinated detection	If this option is selected, outage cells can be detected based on neighboring-cell UE cooperation.
Generate alarm on cell outage	If this option is selected, the MAE-Access generates ALM-311 Cell Outage for any cell that is identified as an outage cell based on the cell KPIs (such as the service setup success rate, abnormal release rate, and RRC setup success rate), or based on a cell outage notification reported by an NE.
Enable long-time no-traffic judgment	The value of No-traffic period threshold ranges from 0 to 720 in the unit of hour. The default value is 96. – If this parameter is set to 0, sleeping-state-based cell outage detection is not enabled. – If this parameter is set to a value other than 0, sleeping-state-based cell outage detection is enabled and the parameter value is configured as the duration for cell outage detection.

2. Click **Preset KPI Thresholds**. A dialog box for parameter settings is displayed. [Table 3-3](#) describes the parameter settings for preset KPI thresholds.

----End

- Activating the function of cell outage detection based on zero access
//Activating the function of cell outage detection based on zero access in SA networking
MOD GNBOAMPARAM: CodrAlgoSwitch=ZERO_ACCESS_DETECT_SW-1;
//Activating the function of cell outage detection based on zero access in NSA networking
MOD GNBOAMPARAM: CodrAlgoSwitch=NSA_ZERO_ACCESS_DETECT_SW-1;
- Activating cell outage detection based on internal faults
MOD GNBOAMPARAM: CodrAlgoSwitch=INTERNAL_FAULT_DETECT_SW-1;
- Activating cell outage detection based on neighboring-cell UE cooperation
MOD GNBOAMPARAM: CodrAlgoSwitch=NCELL_UE_COOP_DETECT_SW-1;
- Activating optimization for high CPRI BER scenarios
SET EQUIPMENT: EQMOPTFEATSW=HIGHCPRIBER_FAULT_POSTPROC_SW-1;
- Activating cell outage detection based on comprehensive features
MOD GNBOAMPARAM: CodrAlgoSwitch=COMPR_FEATURE_DETECT_SW-1;

Self-Healing Policies for Board Component Failures

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Enabling the baseband processing unit exception notification function
MOD GNBOAMPARAM: DevAbnCellUnavblNotifySw=BBP_ABN_NOTIFY_SW-1;
- Configuring the policy for handling board component failures (Take board-level configurations as an example.)
ADD BRDCOMPEXCPCPOL: EFFRANGE=BOARD, CN=0, SRN=0, SN=0, EFFMODE=TIMING;

3.4.1.2 Deactivation

Cell Outage Detection and Recovery

- Step 1** On the MAE-Access, choose **MAE-Access > SON > NR Cell Outage Detection and Recovery**. The cell outage management window is displayed.
- Step 2** Choose **Parameter Settings > Base Station Monitoring**.
- Step 3** In the displayed dialog box, deselect a gNodeB to deactivate the cell outage detection function for the gNodeB. The MAE-Access no longer detects or handles outage cells served by this gNodeB.

----End

- Deactivating the function of cell outage detection based on zero access
//Deactivating the function of cell outage detection based on zero access in SA networking
MOD GNBOAMPARAM: CodrAlgoSwitch=ZERO_ACCESS_DETECT_SW-0;
//Deactivating the function of cell outage detection based on zero access in NSA networking
MOD GNBOAMPARAM: CodrAlgoSwitch=NSA_ZERO_ACCESS_DETECT_SW-0;
- Deactivating cell outage detection based on internal faults
MOD GNBOAMPARAM: CodrAlgoSwitch=INTERNAL_FAULT_DETECT_SW-0;

- Deactivating cell outage detection based on neighboring-cell UE cooperation
MOD GNBOAMPARAM: CodrAlgoSwitch=NCELL_UE_COOP_DETECT_SW-0;
- Deactivating optimization for high CPRI BER scenarios
SET EQUIPMENT: EQMOPTFEATSW=HIGHCPRIBER_FAULT_POSTPROC_SW-0;
- Deactivating cell outage detection based on comprehensive features
MOD GNBOAMPARAM: CodrAlgoSwitch=COMPR_FEATURE_DETECT_SW-0;

Self-Healing Policies for Board Component Failures

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Disabling the baseband processing unit exception notification function
MOD GNBOAMPARAM: DevAbnCellUnavblNotifySw=BBP_ABN_NOTIFY_SW-0;
- Deleting the policy for handling board component failures (Take board-level configurations as an example.)
RMV BRDCOMPEXCPROPOL: EFFRANGE=BOARD, CN=0, SRN=0, SN=0;

3.4.2 Activation Verification

Activation Verification of Cell Outage Detection

To verify the activation of this cell outage detection function, choose **Parameter Settings > Base Station Monitoring** in the cell outage management window, select an NE, and click **Apply**. If a message is displayed, indicating that the NE was added successfully, check whether the function has been enabled for the NE by querying the outage cell list, cell outage alarms, or self-organizing network (SON) logs.

- Verification by querying the outage cell list
If a selected cell is detected as an outage cell, the NE sends the cell outage information to the MAE-Access. The MAE-Access displays all outage cells on the **Outage Cell List** tab page in the **NR Cell Outage Detection and Recovery** window. This indicates that the cell outage detection function has been activated. On the **Outage Cell List** page, choose **More Actions > Reactivate Cell** to reactivate the cell. You can also choose **Reset Base Station** to reset the base station or choose **Reset RRU**, **Reset RRU Through Power-off**, **Reset BBP**, or **Reset BBP Through Power-off** to reset the serving RRU or BBP.

NOTICE

Using the **Reset BBP** and **Reset BBP Through Power-off** methods may affect normal cells because a BBP may also serve a normal cell while serving an outage cell or after deactivation of the outage cell.

- Verification by querying cell outage alarms
If the **Generate alarm on cell outage** check box is selected in the **Parameters Settings** dialog box of the **NR Cell Outage Detection and Recovery** window, cell outage information is recorded in logs and cell outage alarms are reported. To view detailed alarm logs, choose **MAE-Access >**

Monitor > Current Alarm, and then configure the filter criteria by the following operations: (1) Choose **Advanced Settings > Name**. (2) In the displayed dialog box, enter "Cell Outage" in the **Enter a Keyword** text box. (3) Select the **Cell Outage** check box. (4) Click **OK**, and then click **Filter**.

- Verification by querying SON logs
 - a. On the MAE-Access, choose **SON > SON Log**.
 - b. In the left pane of the **Query SON Log** tab page, choose **NR Cell Outage Detection Log** from the **Log Category** drop-down list, and then choose **NR Cell Outage**. (There are no requirements for the other settings in this pane.)
 - c. Click **Query**. If cell outage logs are listed, cell outage detection has been activated.

In addition to the preceding three observation methods, you can also use the following methods to check whether the cell outage detection function has taken effect:

- Cell outage detection based on zero access
You can run the **LST NRCELLOAMPARAM** command to check whether the function of cell outage detection based on zero access has taken effect. [Table 3-17](#) describes the parameters for verification. After the function of cell outage detection based on zero access is enabled, the gNodeB calculates busy hours and an adaptive threshold based on historical cell data. The values of parameters listed in [Table 3-17](#) are valid only after this function has been enabled for at least one week. (Non-default values indicate valid values.). Until then, this function has taken effect.

Table 3-17 Parameters related to cell outage detection based on zero access

Parameter Name	Parameter Description
Busy Hour Start Time	Indicates the start time of busy hours for zero access detection. The value of this parameter is automatically calculated by the system based on the busy-hour traffic of the cell. The default value is 255 .
Busy Hour End Time	Indicates the end time of busy hours for zero access detection. The value of this parameter is automatically calculated by the system based on the busy-hour traffic of the cell. The default value is 255 .
SA Msg1 Num Threshold	Indicates the threshold expressed as the number of Msg1 messages for zero access detection. If the number of Msg1 messages is greater than this threshold, zero access detection is performed. The value of this parameter is automatically derived from training based on the number of Msg1 messages (N.RA.Contention.Att) in busy hours. The default value is 4294967295 .

Parameter Name	Parameter Description
SA Msg2 Num Threshold	Indicates the threshold expressed as the number of Msg2 messages for zero access detection. If the number of Msg2 messages is greater than this threshold, zero access detection is performed. The value of this parameter is automatically derived from training based on the number of Msg2 messages (N.RA.Contention.Resp) in busy hours. The default value is 4294967295 .
SA Msg3 Num Threshold	Indicates the threshold expressed as the number of Msg3 messages for zero access detection. If the number of Msg3 messages is greater than this threshold, zero access detection is performed. The value of this parameter is automatically derived from training based on the number of Msg3 messages (N.RRC.SetupReq.Att) in busy hours. The default value is 4294967295 .
NSA SgNB Access Num Threshold	Indicates the threshold expressed as the number of SgNB additions for zero access detection in NSA networking. If the number of SgNB additions is greater than this threshold, zero access detection is performed. The value of this parameter is automatically obtained by the system based on busy-hour training data of the cell. The default value is 4294967295 .

- Cell outage detection based on internal faults
You can run the **DSP NRDUCELLTRP** command to query the fault pre-detection result of downlink transmit power for the function of cell outage detection based on internal faults. The base station performs cell outage detection based on internal faults only when the downlink transmit power fault pre-detection result is "passed". If the downlink transmit power is 0 and no UE is performing downlink data services, and the fault duration meets the related requirements, ALM-29875 No Traffic on NR DU Cell TRP is reported. (The fault duration threshold for triggering alarm reporting is specified by the **gNBOamParam.SlcAlmDetectDurationThld** parameter. The recommended value is **24**.)
- Cell outage detection based on neighboring-cell UE cooperation
You can run the **DSP GNBOAMPARAM** command to query the latest cell-level detection result. If a cell is detected as a suspected outage cell and a neighboring-cell UE attempts to hand over to it, the detection result of the cell is displayed. The cell outage detection based on neighboring-cell UE cooperation function is performed only when the cell's detection result can be queried.
- Cell outage detection based on comprehensive features

You can run the **DSP GNBOAMPARAM** command to display the activation status of cell outage detection based on comprehensive features. If the detection result of the local cell is available and the value of **Compr Feature COD Activation Status** is **FUNCTION_IN_EFFECT** for the local cell in the command output, cell outage detection based on comprehensive features has been successfully activated.

Measurement Period Check

To check the measurement period for cell outage detection based on abnormal preset KPIs, perform the following steps on the MAE-Access:

On the MAE-Access, choose **MAE-Access > Performance > Measurement Management > BTS5900 5G > gNodeB Function > Measurements Related to Algorithm(5G) > DC Measurement**. In the right pane, click the **Status** tab. Check the value in the **Period** column. You can also click names of different base stations in the lower left corner to check the measurement periods for different base stations. Cell outage detection based on abnormal preset KPIs uses the shortest performance measurement period (other than 5 minutes) available in the **Measurement Management** window on the MAE-Access.

3.4.3 Network Monitoring

None

4 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*